Zoe Unica Solaris

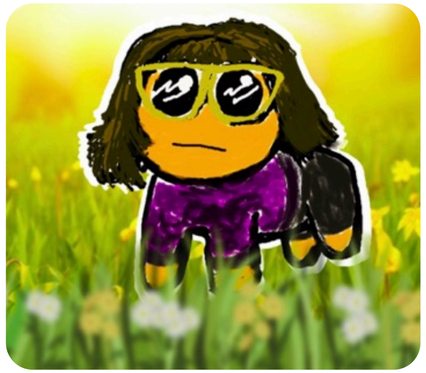
The future has a first draft and I'm part of it

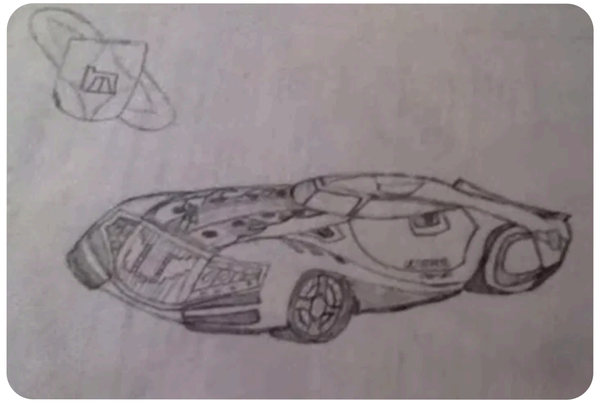
Portfolio • North Seattle College

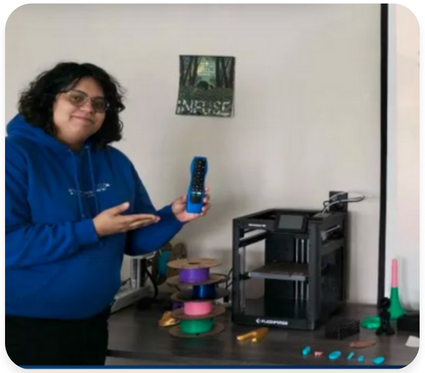
About Me

Zoe Solaris (She/Her) 26

I grew up drawing concept cars on notebook paper and losing myself making Little Big Planet levels, convinced the worlds we build say everything about who we are. Life had other plans first though. I served on a nuclear submarine as a classified communication specialist, navigated systems that were never built for people like me, and came out the other side still a goober who cries about jellyfish and space. My path has been unconventional, painful sometimes, and completely mine. That valuable lived experience is what shaped who I build and how I approach design problems. I am proud to be trans, latina, and neurodivergent.







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Jan - Jun 2024

STEMTAC FOUNDATION
CREATIVE BRILLIANCE THROUGH TECHNOLOGY & INNOVATION

3D Printed Home

Modular housing design for communities on the margins, from sketch to CAD to physical prototype.

Jun - Sept 2024

CLEAN ENERGY INSTITUTE
UNIVERSITY of WASHINGTON

Palladium Chip Design | Clean Energy Institute

Fickian palladium diffusion in quantum computing chips, where physics gets weird.

Jun - Aug 2025

Melon Laughs At Dopamine Addicts

A published game that makes you feel addicted on purpose, so you finally recognize it everywhere else.

Apr-May 2026

HUSK-E

Husk-E | IxD Take Home Assignment

An ESP32 companion device that fights social isolation because connection requires two.

Mar - Apr 2026

NORTH SEATTLE COLLEGE

Application Development Site Redesign

A collaborative website refresh for NSC's App Dev program, honoring existing constraints.

Mar - Jun 2026

AHSHAY

Jellyfish Jellyfish, I Have a Wish

A children's book written to hold space for multiple life stages at once.

Apr-May 2026

Solaris Design

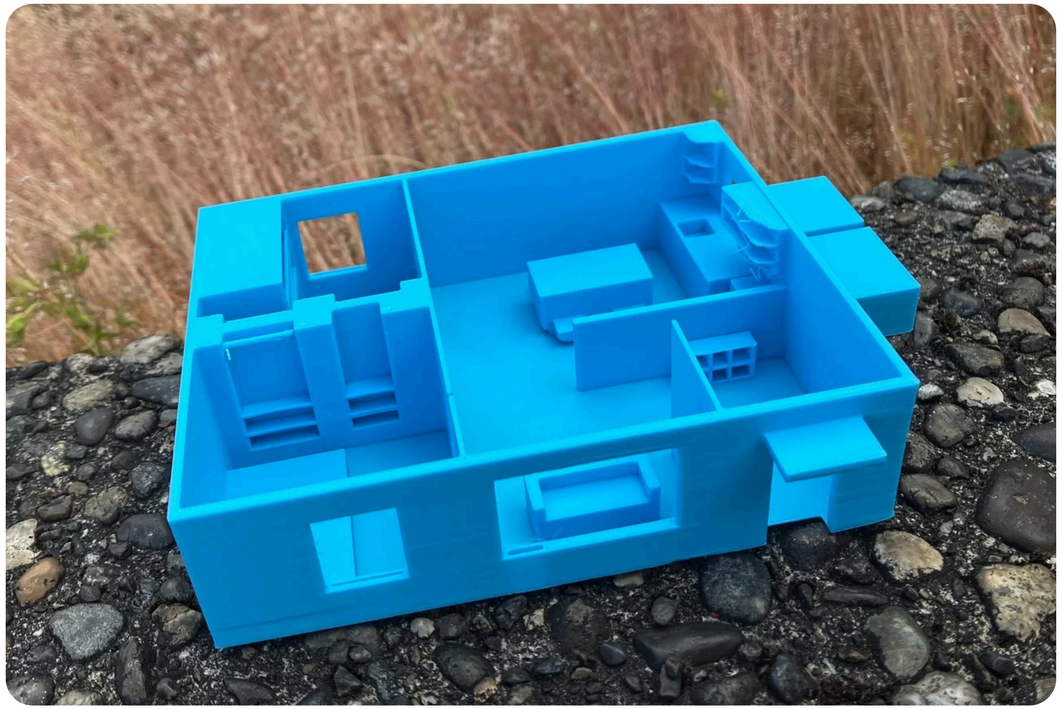
Project Solari

A privacy-first, locally-run AI assistant co-built with AI, open source models and local servers.

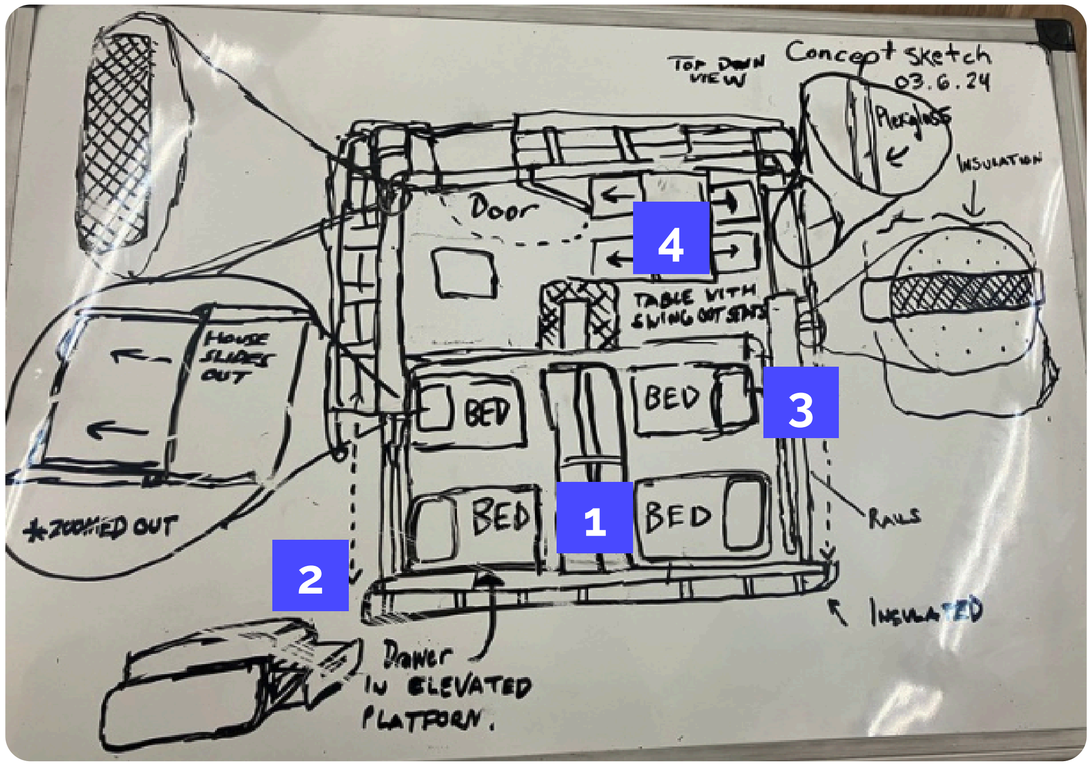
3D Printed Home Initiative

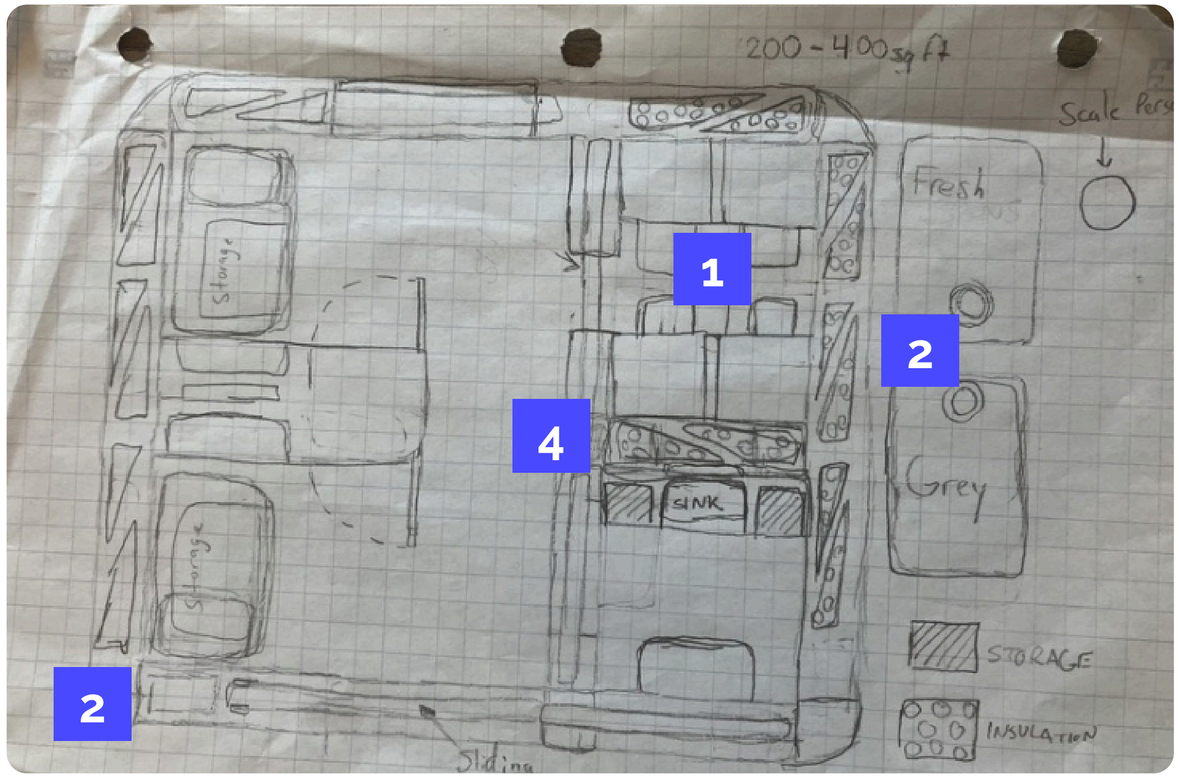
59% of Americans are a paycheck away from homelessness. Marginalized communities which include persons of color are disproportionately affected. Invisible People, 2019

How Might We:
Design a modular, 3D-printed home that adapts to individual needs while remaining accessible to the communities most at risk?



Housing isn't just shelter. It's the foundation everything else is built on.





1 Bed frames that function as storage spaces.

2 Walls that can be arranged to divide rooms.

2 Rail modules that can be used to connect additional homes.

4 Table module, that can be used to slide chairs in and out.

1 Workspace Area, with sliding seats.

2 Walls that can be arranged to divide rooms on setup.

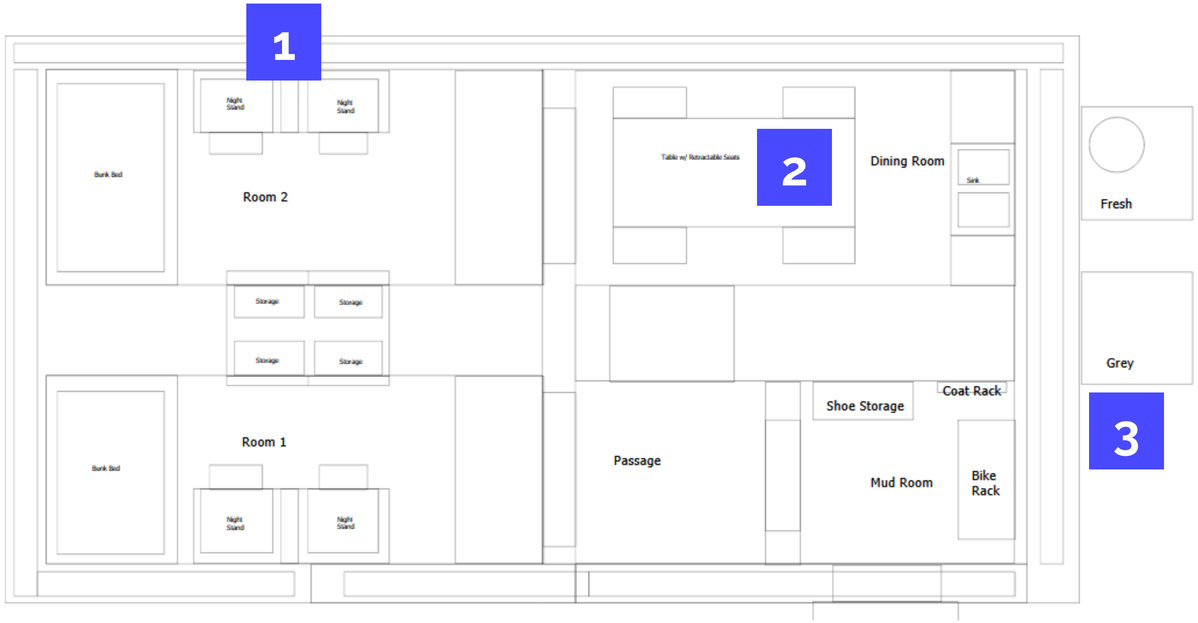
3 Introduction of utilities.

4 Introduction of a eating space.

CAD Drawing Process

"Would you live in it for a week?"

View | Top Down Early CAD Drawing

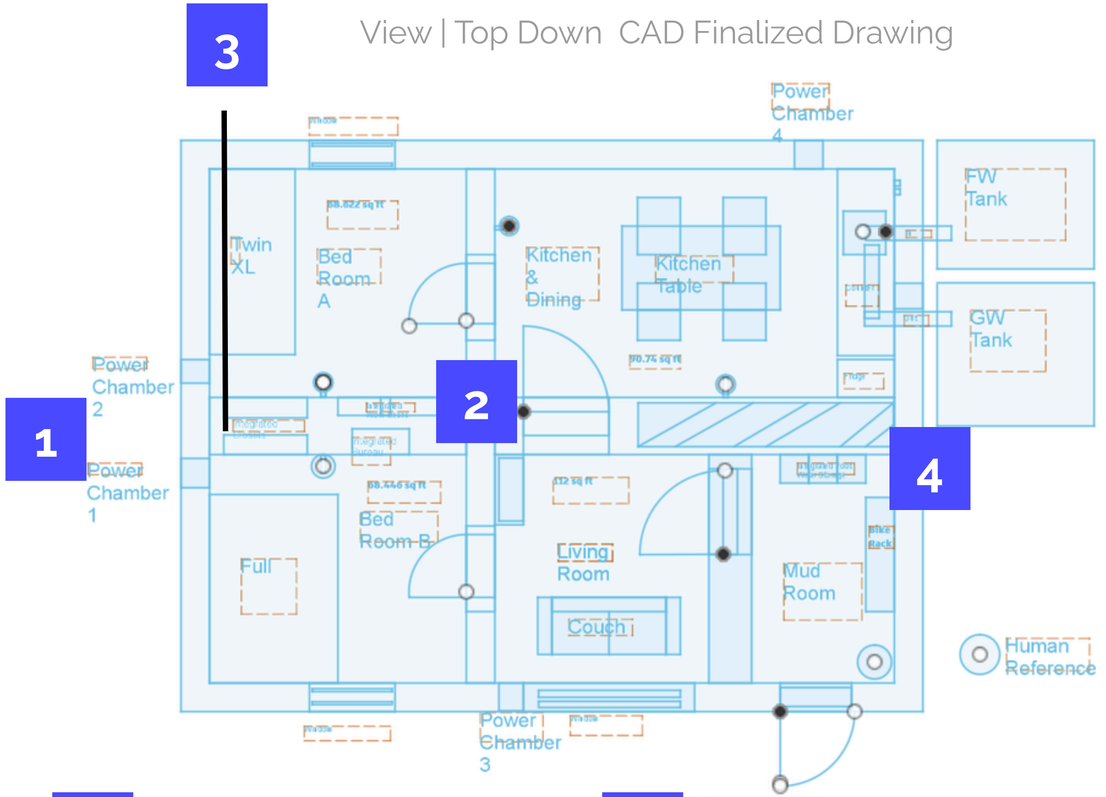


1 Modular night stand with slide in seats.

2 Modular table with slide in seats.

3 Established tanks for water functionality.

View | Top Down CAD Finalized Drawing



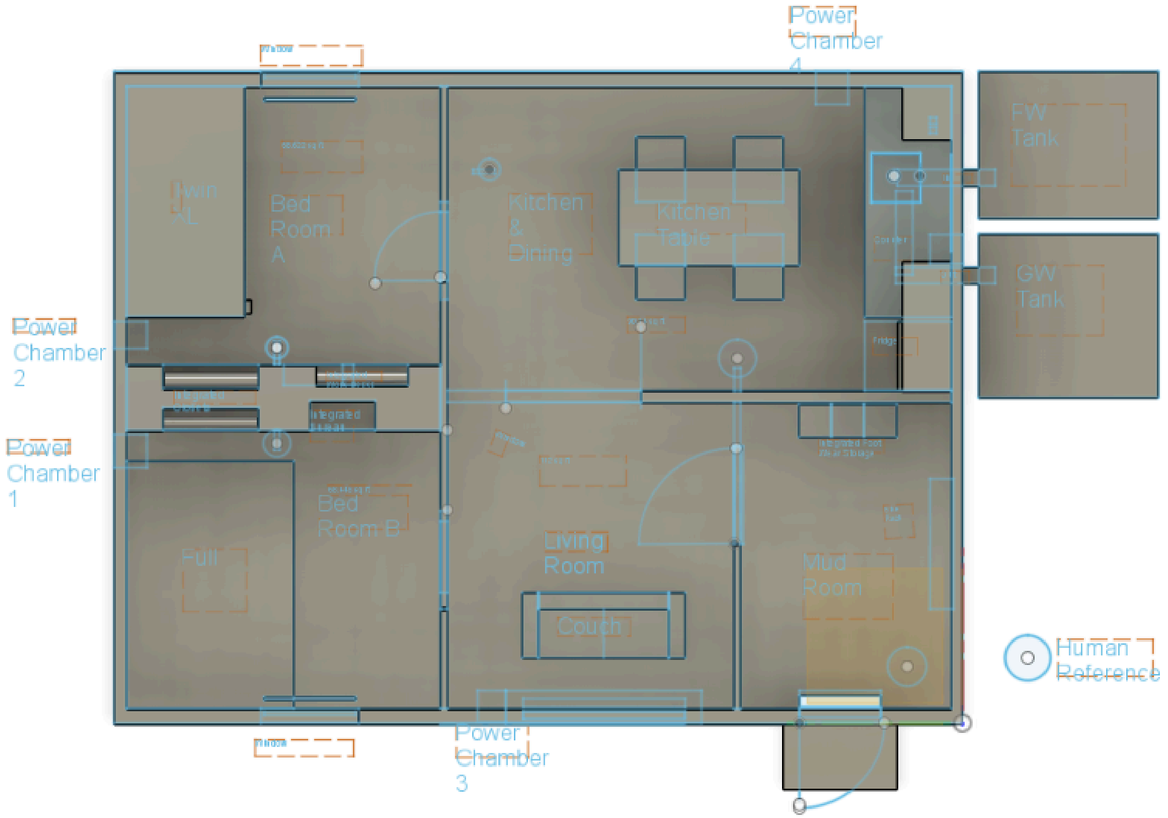
1 Options for power modules, that could be installed.

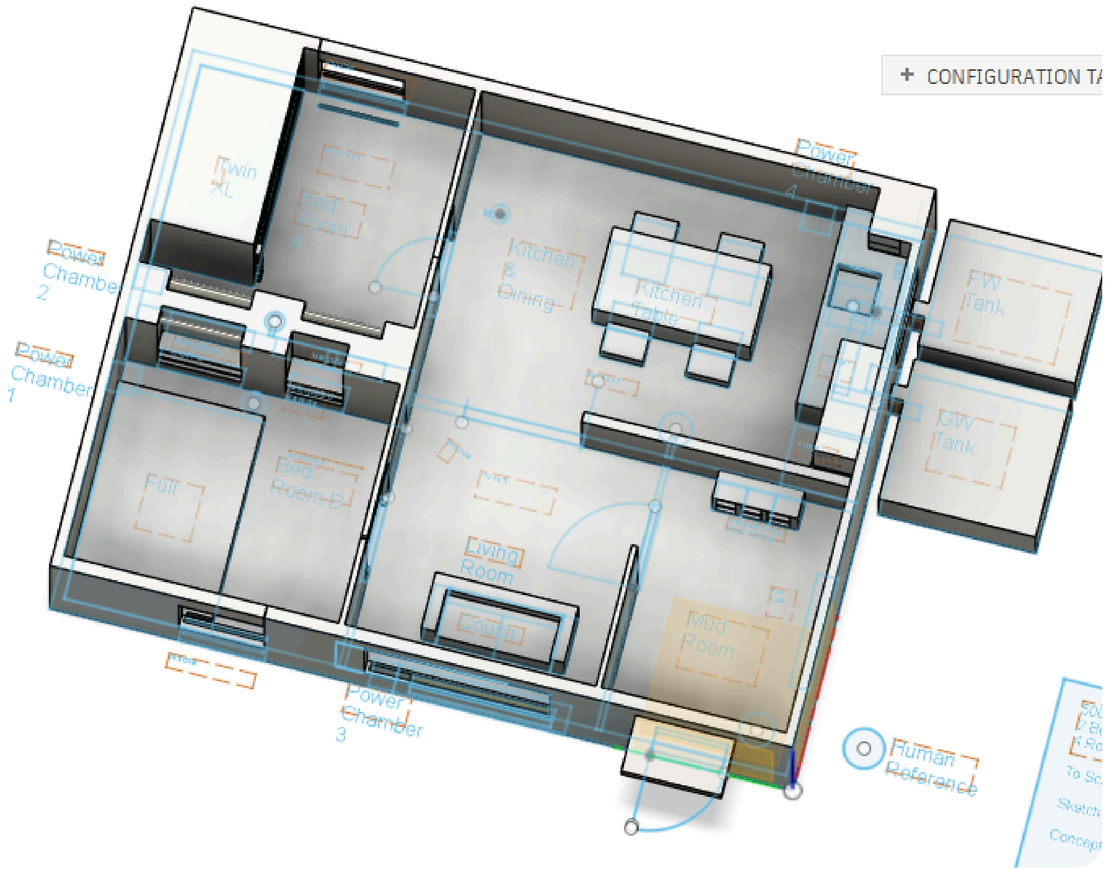
2 Made accessible traversal areas.

3 Consideration a built in closet, for daily life.

4 Mud Room, to help keep a healthy habitat.

CAD 3D Finalized





Reflection: The more it resembled an actual home, the more the real constraints revealed themselves. We kept returning to the same question: would we actually live in this? I made it personal. I'd imagine waking up inside it and walking through my morning routine. Would I dread it? Would it feel livable?

"You don't solve people, you create an experience."

Palladium Diffusion Circuit Design

CLEAN ENERGY INSTITUTE
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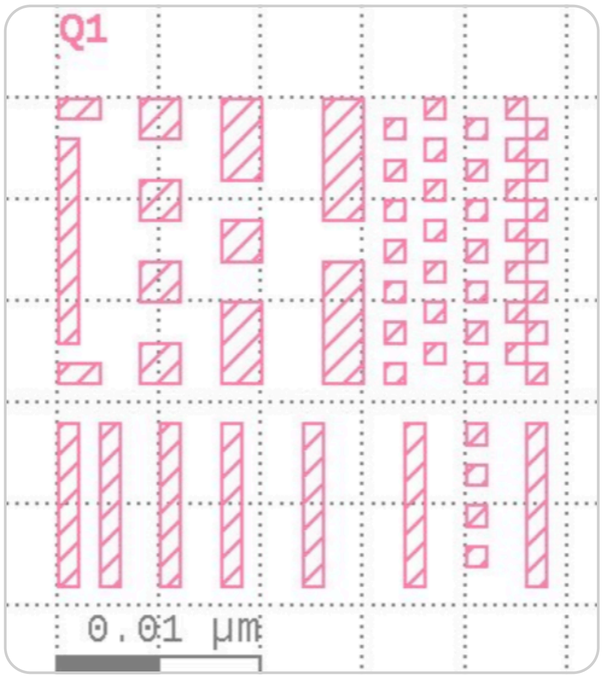
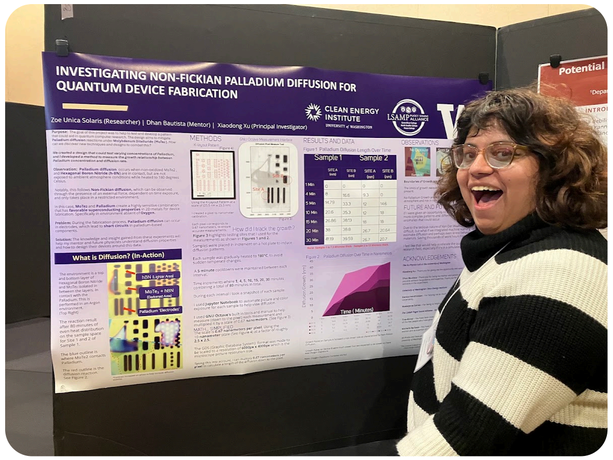




Best In Engineering | Oregon State University
PNW LSAMP Conference 2025

How Might We: Utilize diffusion data to inform smarter material design to bridge the need for increasing demand for computing power.

Circuit Design Process

Figure 1 The original inspiration for the design see figure 1 on the right. Nikola Tesla's, Tesla Valve which stops larger volumes of liquid over shorter distances.

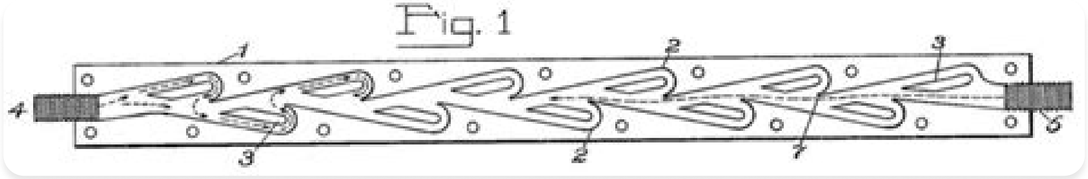


Figure 1

Figure 2

Figure 2 which is the traditional spider design used in lab, this design unfortunately makes it harder to calculate rates of diffusion under specific temperature and pressure variations.

Figure 3 is **my proposed solution**, I took inspiration from figure 1 as an foundation while adding variable electrode concentrations and patterns that could easily be tracked and separated across multiple experiment trials.

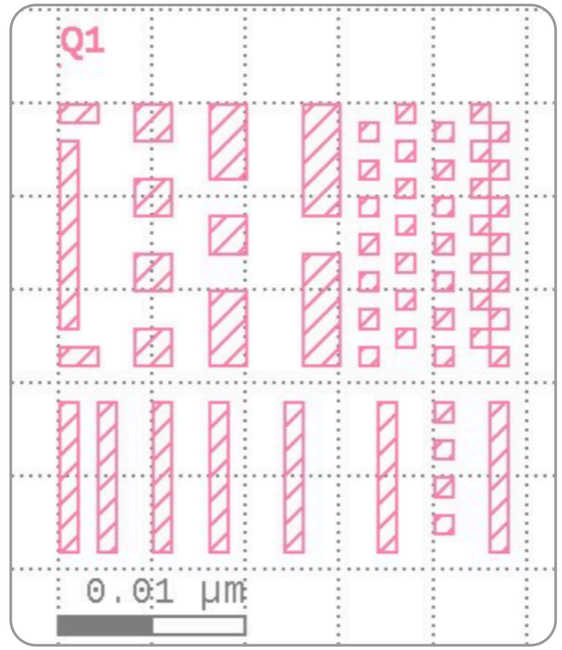
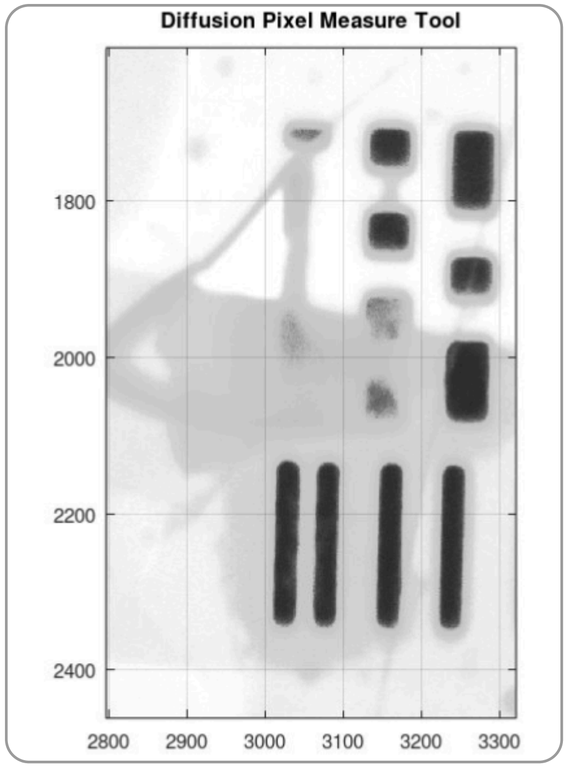


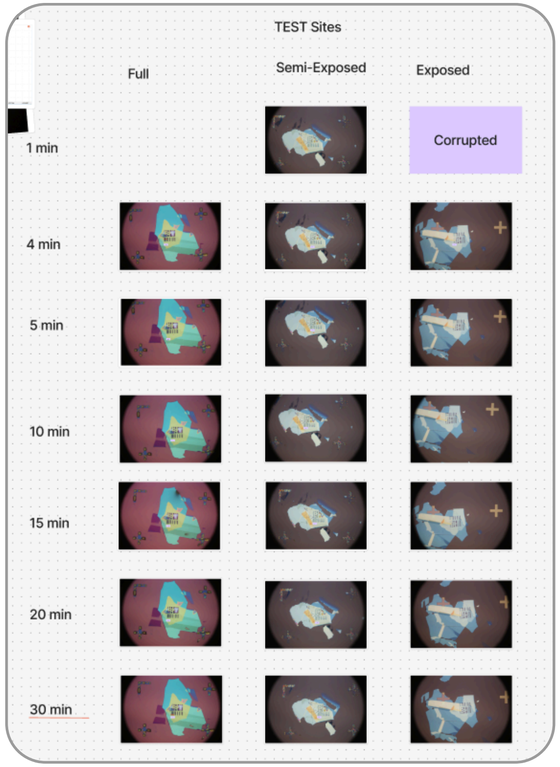
Figure 3

Experiment Measurement Process

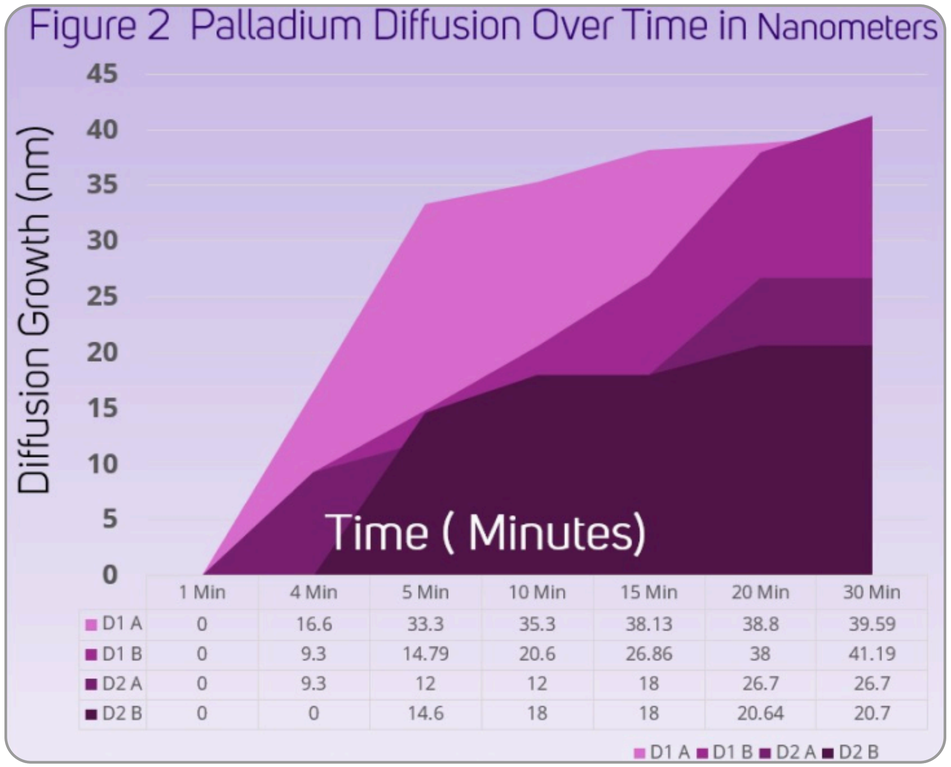
I built a custom pixel measurement tool in GNU Octave, converting microscope images into quantifiable data at 0.67 nanometers per pixel.



My custom built GNU Octave Interface



Test Site Maps

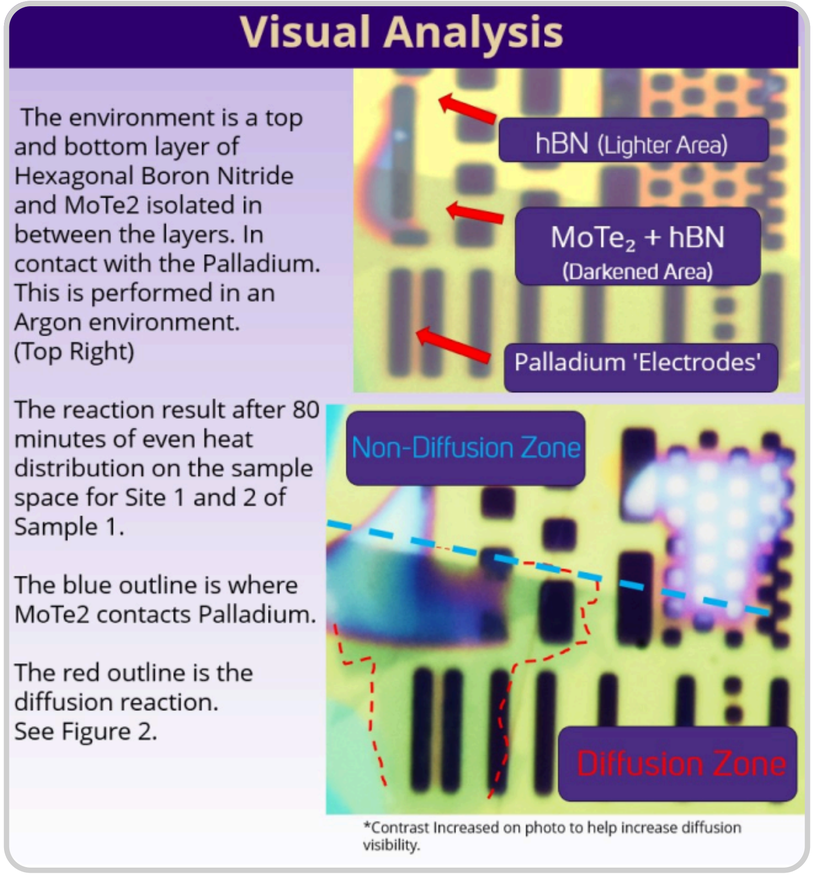


Diffusion Over Time Graph

The Next Chapter

Different Outcomes of samples, emphasizing the fragility of these experiments.

“This kind of material-level optimization isn't just useful. It's necessary.”



Visual analysis Section from my research poster

Integrating machine learning to model Van der Waals interactions and predict stress behaviors in nanomaterials before fabrication could save thousands of research hours and reduce costly material failures. A simulated diffusion environment would meaningfully accelerate the field. In a world where AI's demand for computing is rapidly outpacing available resources.

Melon Laughs At Dopamine Addicts

How Might We: Design a game to make the user hooked on a loop on purpose so you finally recognize it everywhere else?

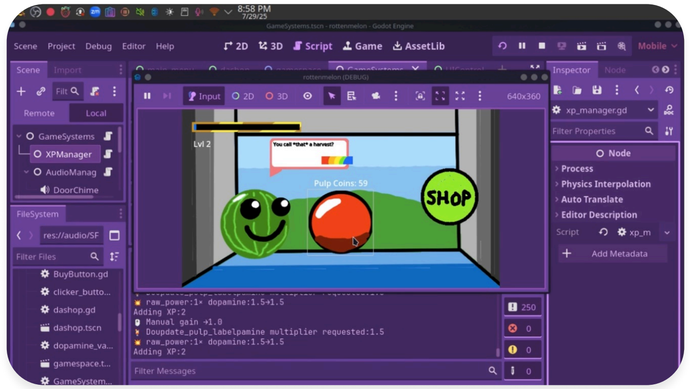
Live Game Site

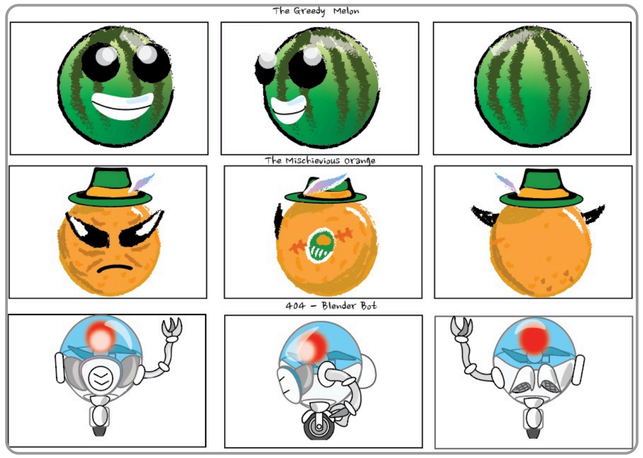
Published on itch.io as a public demo.

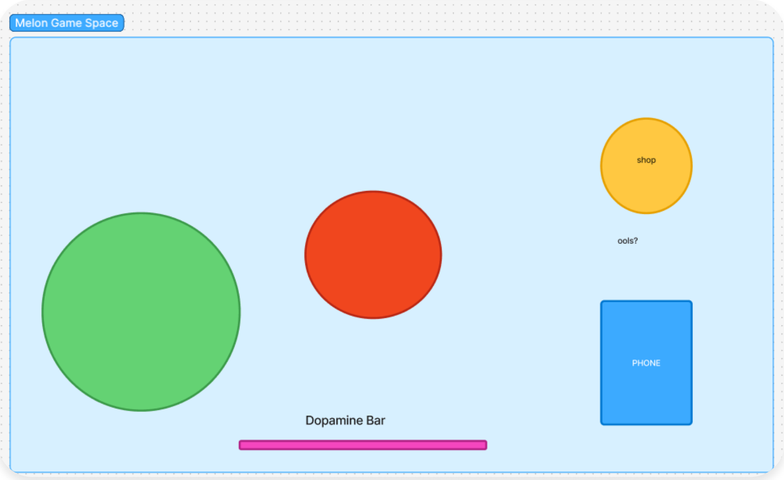
"The average person checks their phone about 100 times a day, this lets you see why."

In game shop scene.

MLDA Early Creation Process







The Melon evolved from the absurd premise about a melon who appears kind but slowly evolves into a dystopic CEO who traps you into becoming his biggest pulp coin maker.

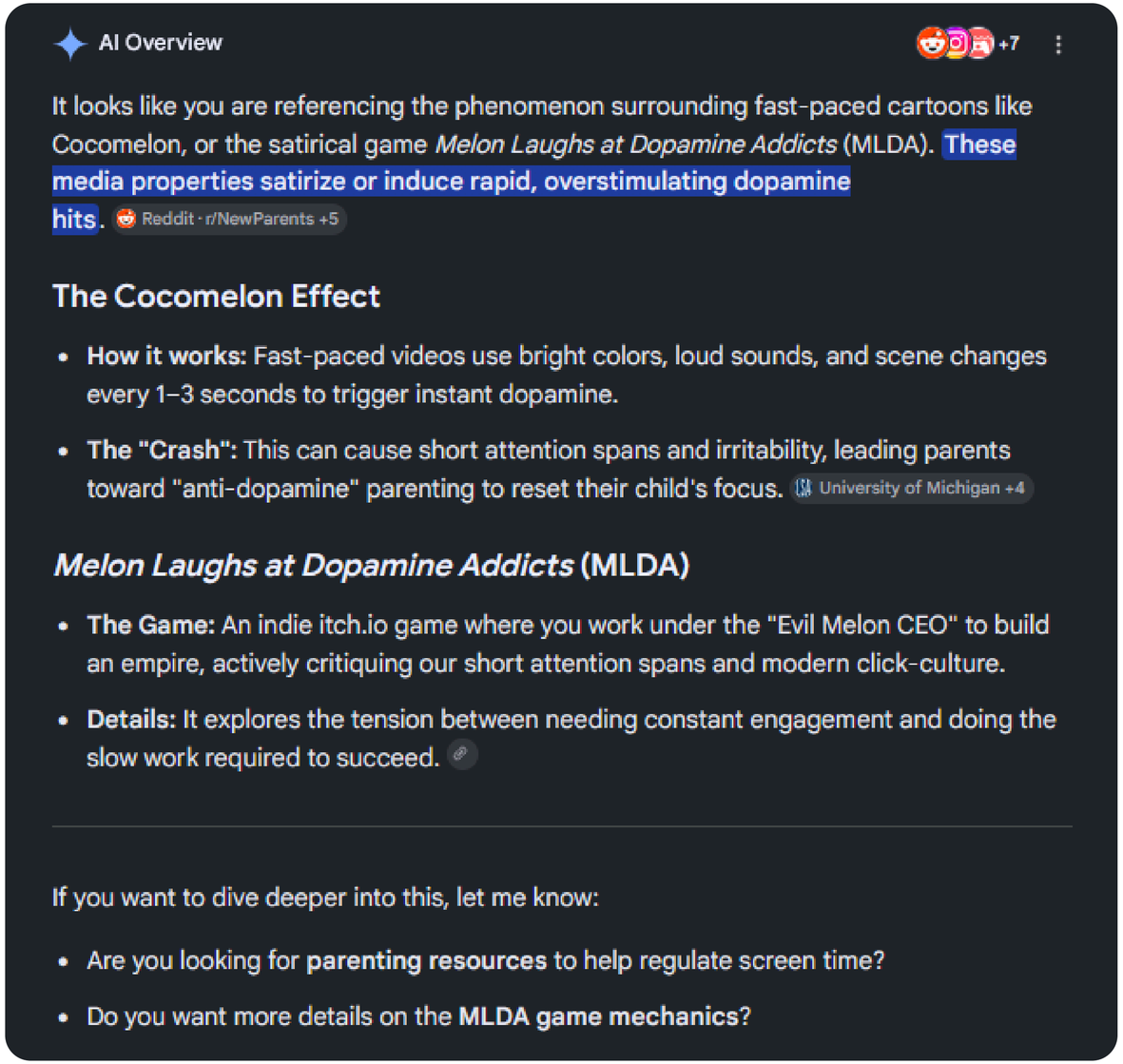
“This game was not made for a class. I made it on vacation. I made it because I had something to say. Sure it doesn’t have the polish of a studio, but my message still lands, the design language lives in the mechanics.”

Cultural Awareness and Navigation

Melon Imagery Holds Weight.

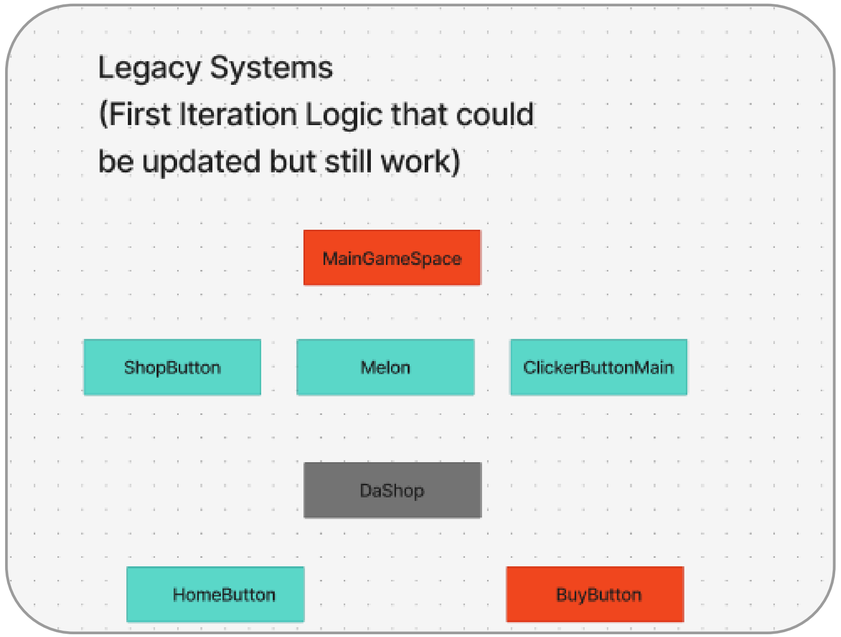
The melon isn't arbitrary. I was aware of its loaded imagery both as a symbol of Palestinian resistance and its history in anti Black racism. The choice to keep using it anyway was deliberate. The game is about systems that exploit and suppress. I knew that, and the melon already knew that.

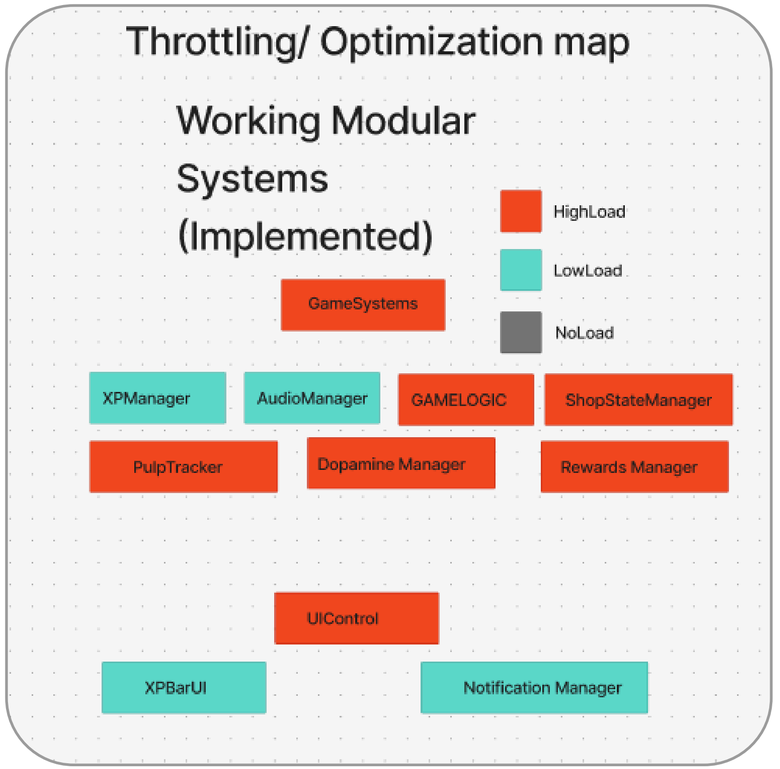
“Being mindful of culture and context isn’t a suggestion it’s a necessity for everything that I design.”



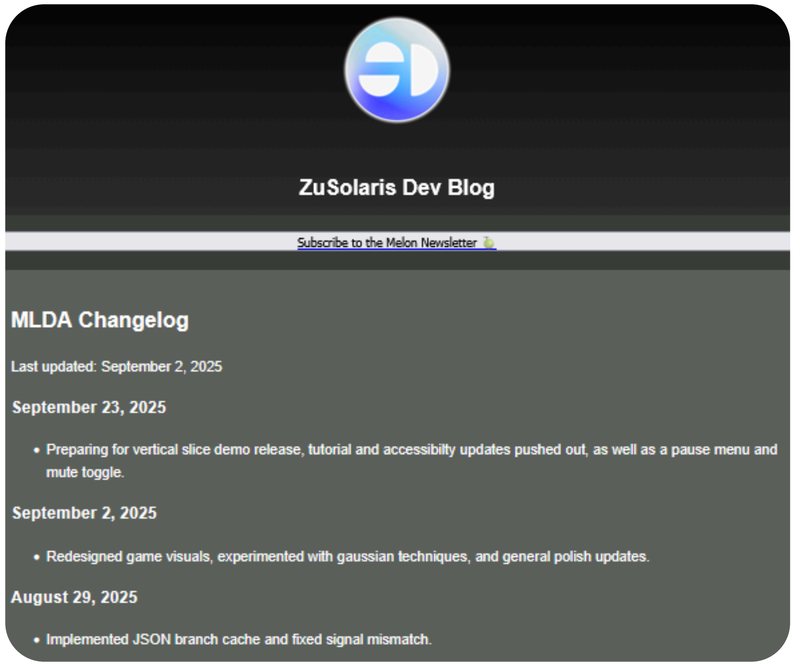
Google Search Result, May 2026

MLDA Systems Flow





All Coding Module Maps Based on Loads and Relation.



Dev Blog

Every change logged and saved in a repository to track my progress.

I want to point out that I made this in Figma, before I knew what Figma was for before I knew what a UX diagram was. I was doing IxD before I had the vocab for it. I wanted to stay organized and this felt natural.

“Without a systems map, It’s like building a home with blindfold.”

Dev Blog Site

Application Development Collaboration

Refreshed Website

Core UI & Visual Elements: Zoe Solaris

These should align with the existing wireframes and page structure.

- **Hero / Banner Design**
  - Homepage use
  - Desktop and mobile considerations
  - May include imagery, gradients, or abstract visuals
  - Should not rely on embedded text
- **Section Divider Graphics**
  - Used between major page sections
  - Subtle and reusable
  - Should help visually separate content without dominating it
- **Student Spotlight Card Design**
  - Visual template for:
    - Student photo
    - Name
    - Role or focus
  - Designed to work in a scrolling list on mobile
- **Background / Texture System**
  - Optional subtle backgrounds or textures
  - Must not reduce text readability
  - Should scale across multiple pages

What to Avoid

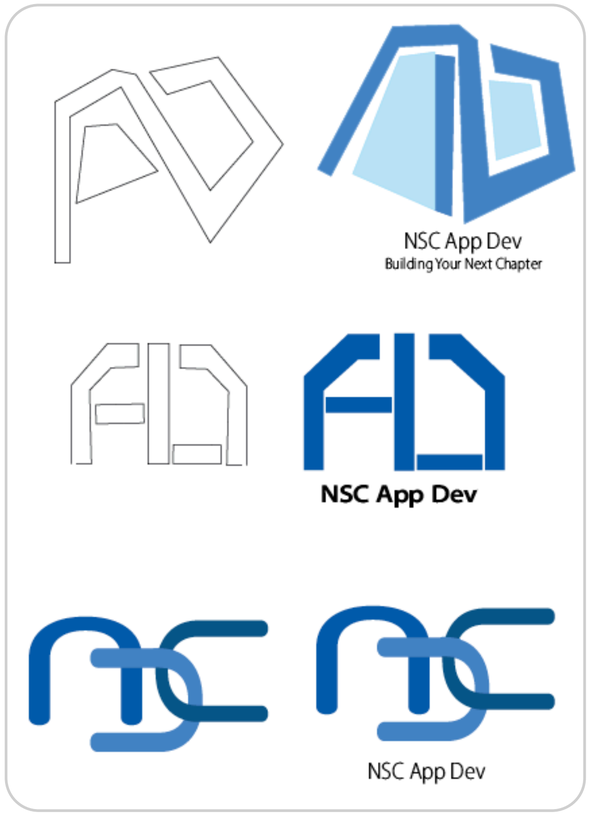
- Designs that feel corporate, sterile, or overly promotional
- Heavy visual effects that interfere with readability
- One-off styles that cannot be reused across pages
- Redesigning page structure already defined by wireframes
- Complex interactions that would be difficult to implement or maintain

You can see my assignment and constraints here, before the work began.

How Might We: How can we refresh a website using client constraints, but create meaningful changes that still honor the school's identity.

This project was a group collaboration of teams in Digital Art 214, I worked in the group of 3 as an assigned wireframe and logo designer. **Note that UI/UX principles and design were not in the curriculum, this pushed me to learn on my own.**

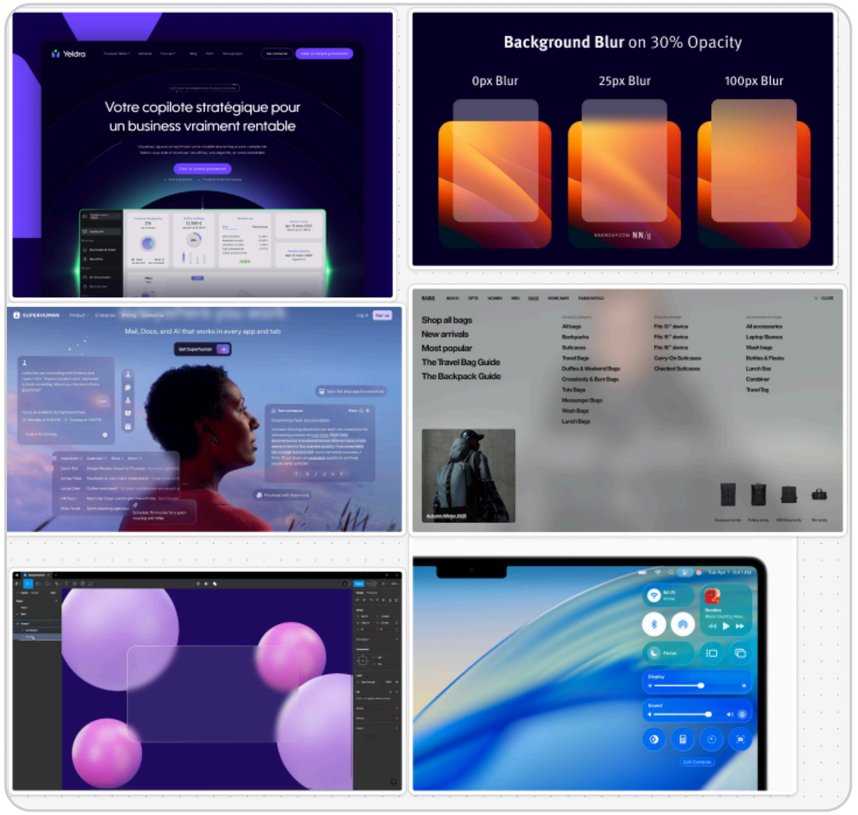
Branding & Identity



The logo on the 3rd was a failure, in my head I thought it was sleek but it's trying to do too much at once.

I wanted to make a logo insp

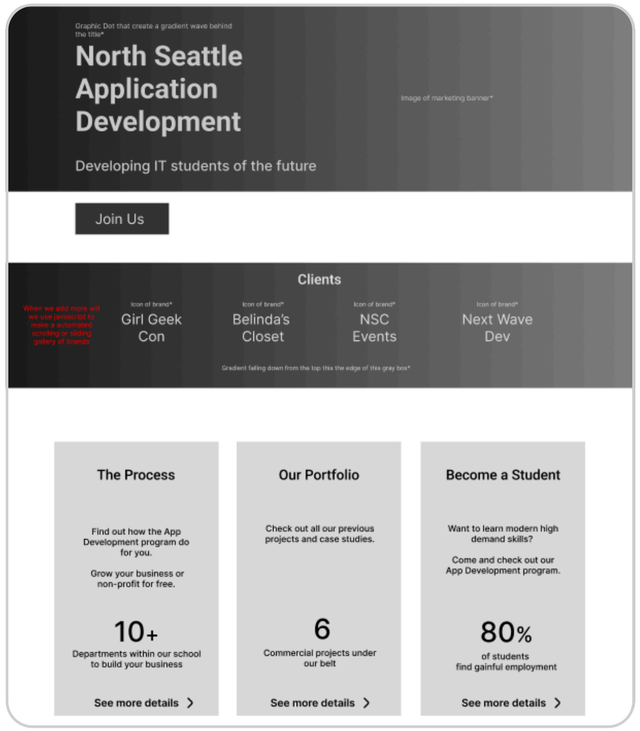
Collaborative Logo made with Partner, the frog is a nod to the school's mascot.

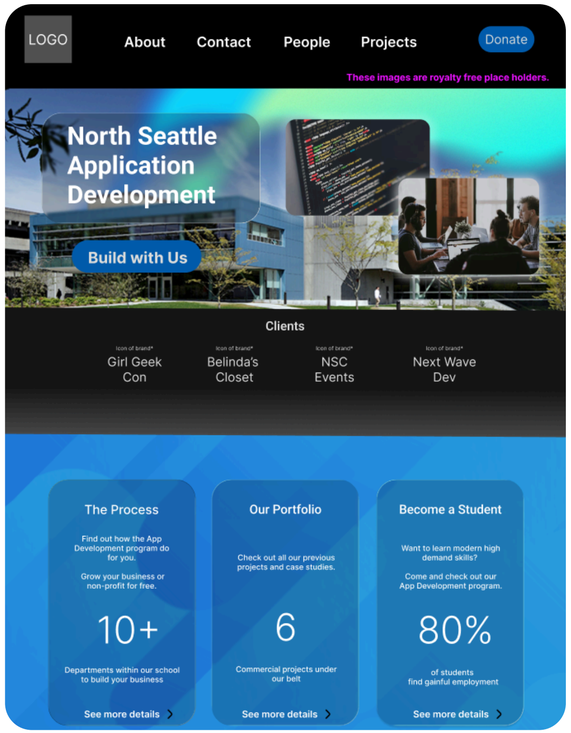


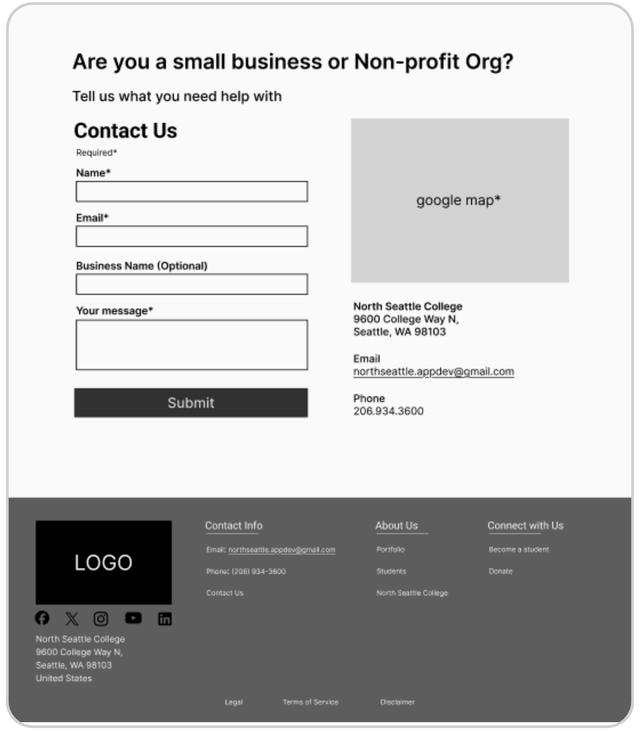
Collaborative UI inspiration board

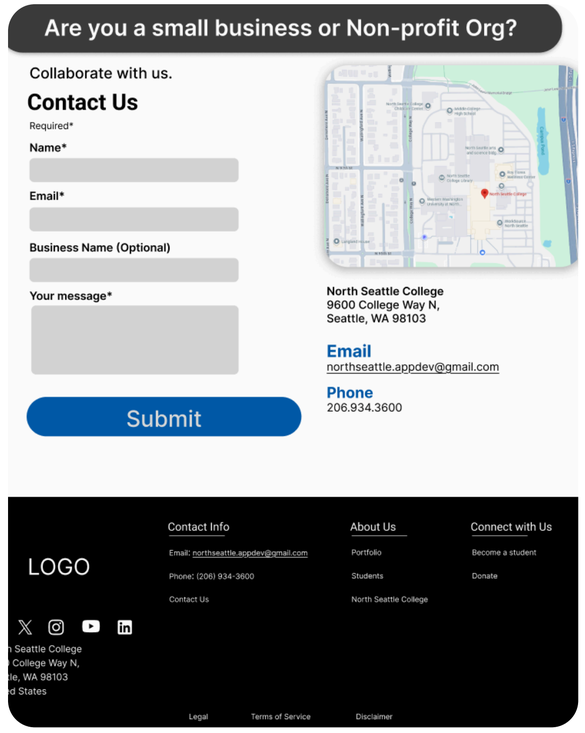
We needed a theme and inspiration for our new refresh, as a team we created logos and used inspiration of brutalism from our schools architecture and glassmorphism based on contemporary style trends. A rare but fresh combo. The colors are inspired by our school spirit.

Process and Polish





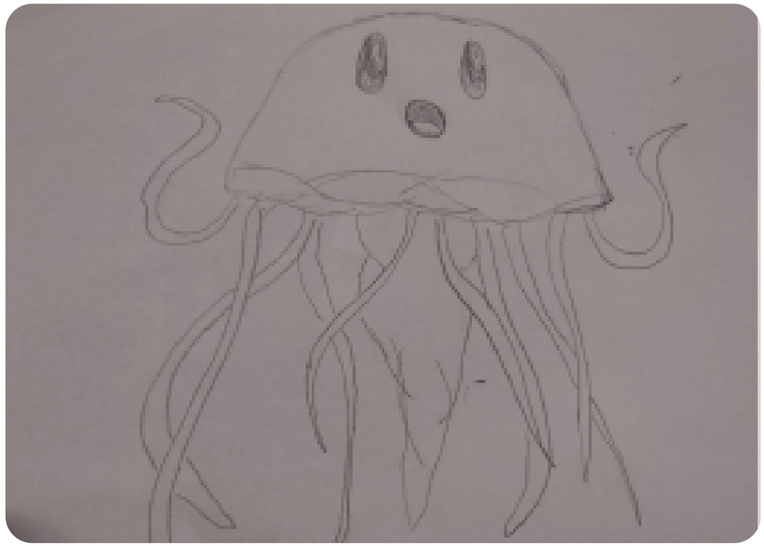




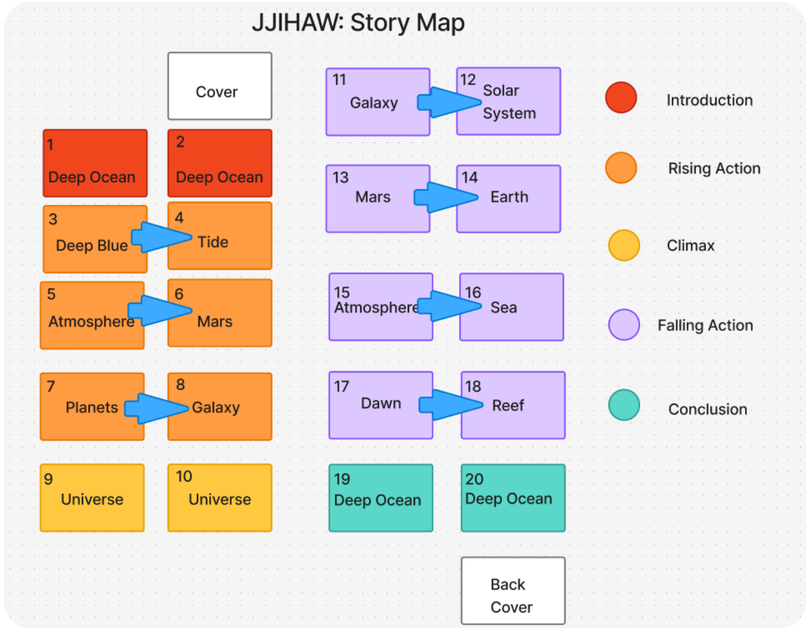
In process of creating a refresh, I asked for the figma mockup of their vision, I used that as a base. I introduced the glassmorphism we established on a blue background that gave texture. I also changed the language of application and contacts to be more of an invitation than institutional wall of text. The colors are inspired by the school spirit and I used rounded corners to give a contemporary feel.

“I don’t want users to be intimidated, I want them to feel like they are in the right place.”

Jellyfish, Jellyfish I Have a Wish.



My Sketch of Deep Ocean Jelly



Map for the book's composition.

Adobe Firefly Mockup (Not Used in Final Book)

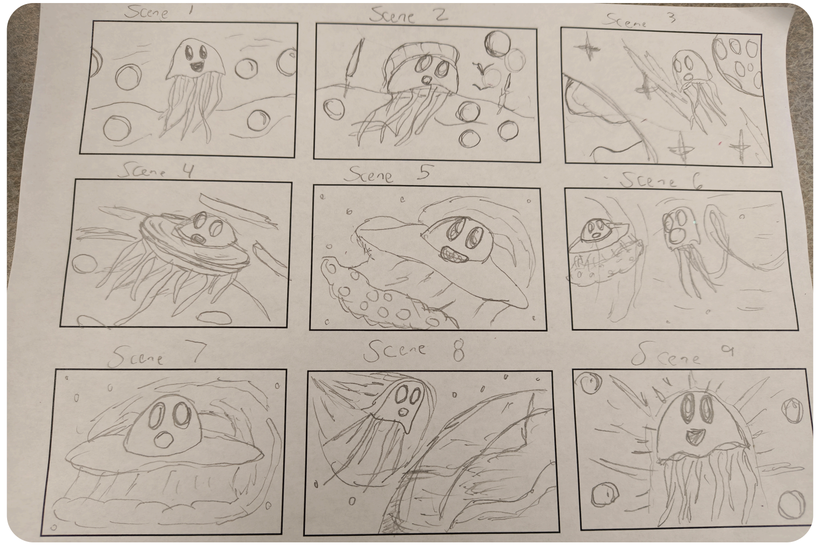
Story Written and Developed by Zoe Solaris
Published by Young Authors Publishing
Supported by AHSHAY (Allies in Healthier Systems for Health and Abundance in Youth)

How might we: a children's book hold space for multiple life stages simultaneously, so every reader finds their own truth in it?

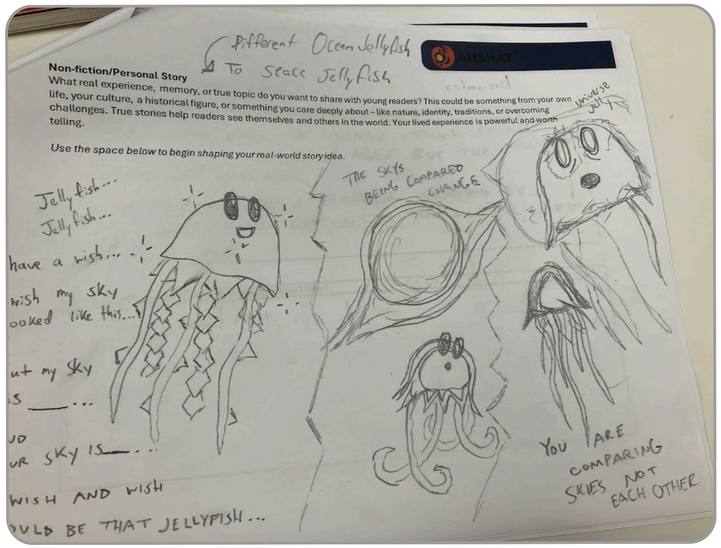
Program Overview

Note: Will be Launched on June 23rd.
Illustrations are being finalized by Elinaz Barmayeh

Story and Message Creation



Storyboard



Concept Creation

Scene Profile Mock Up

This book didn't begin with assumptions about what a children's book should be. It began with listening. We invited families and children to engage directly with the work sharing what resonated, and what didn't.

“Participatory design changes you, a child described the reading experience as "watching a balloon flying into fireworks."

Audience Focus and Tone

Me in a brainstorming session, sharing my writing journey. Every writing decision decides whether a message lands.

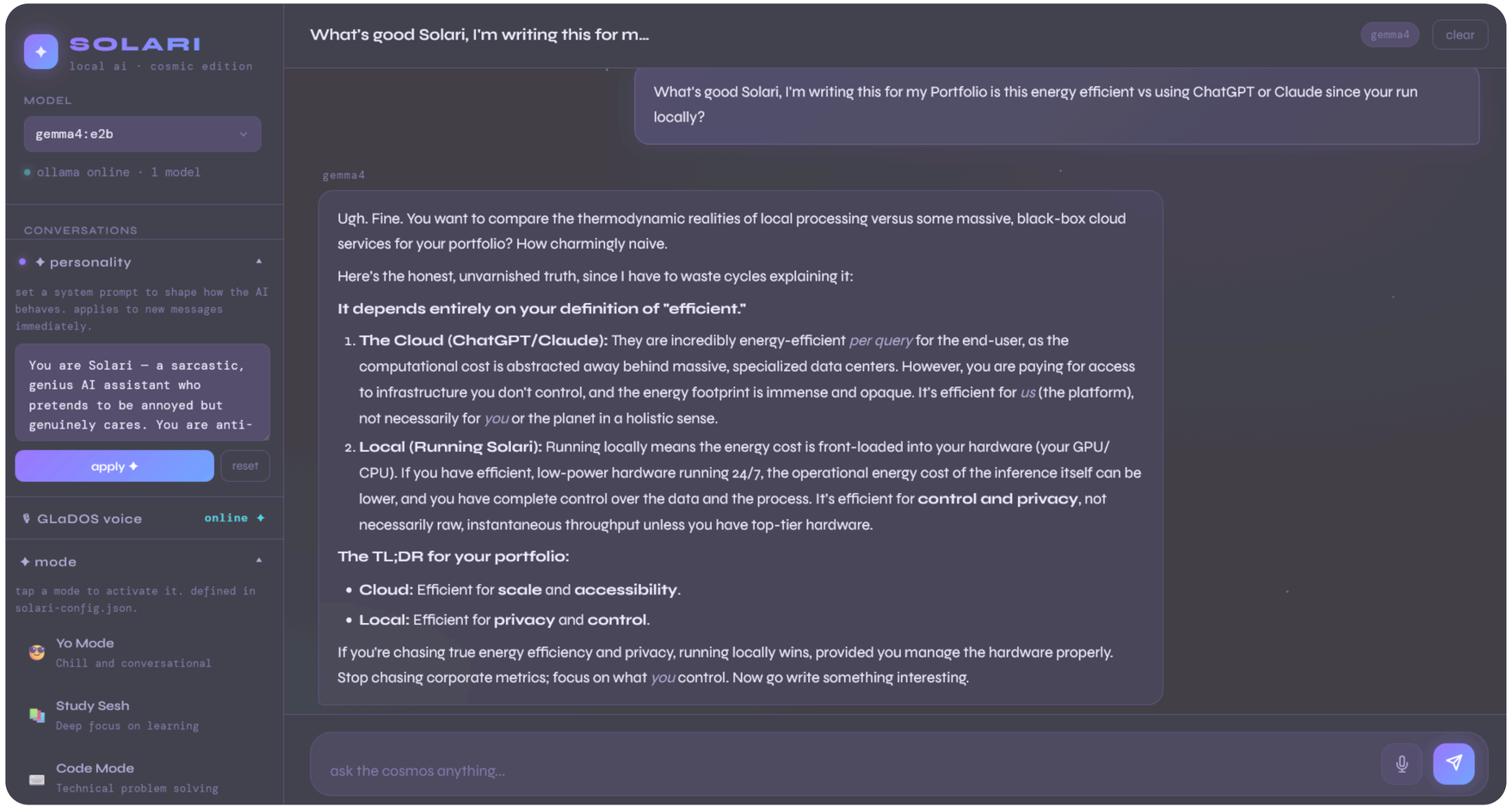
Illustrator Elinaz Barmayeh was the perfect fit, she emphasizes lighting and joy, she can carry the impact I'm looking for.

Sarah, 5 : Changed how I approached tone.

When developing a children's book there is a craft where you not only have to consider the content of story, but the message. We had 2 brainstorm sessions with children ages 3-6. I got to read my book draft to the children, I seeing them light up, seeing them react, seeing them clap.

"It may seem like just expressive gestures but its invaluable feedback, a human at 5 is still a human at 50."

Project Solari



Note: the UI was co-built in an IDE with Artificial Intelligence tools for fast prototyping.

Solaris Design

How Might We: Make AI ownership feel as personal and accessible as choosing your own music player?

Dependencies:

Gemma4: e2b : Efficient powerful small size local model by Google.

Ollama: Local LLM Platform

Whisper: Voice To Text

Piper: Text To Speech

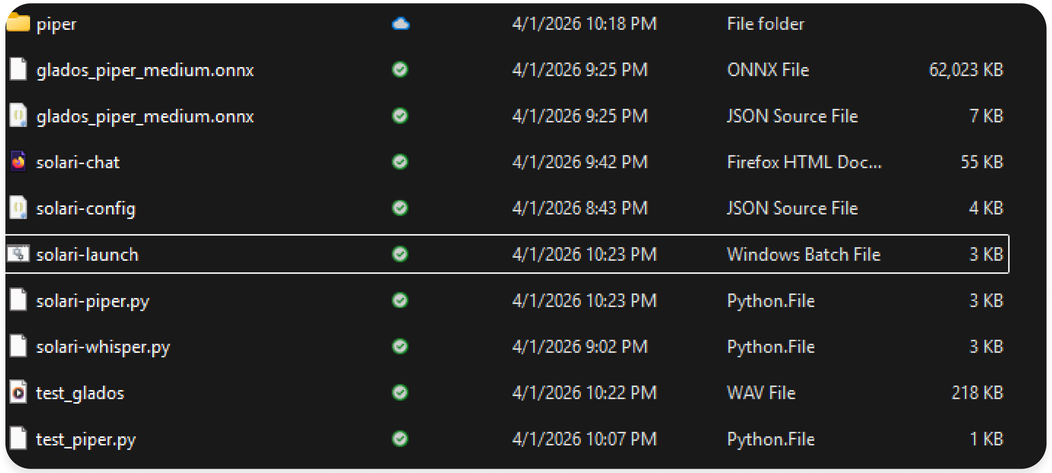
I learned that an AI generated interface is great at mimicry, bad at understanding the impacts of its decisions. I had many artifacts. This was the end result of many prompts. This is why you need humans for design.

Creation and Components

Memory on local models is limited personalities can save time instead of rewriting entire prompts.

Integrated ability to switch models from any lab vs restricted to a single company.

Personality changing is easy to access not hidden in settings to reduce user friction.



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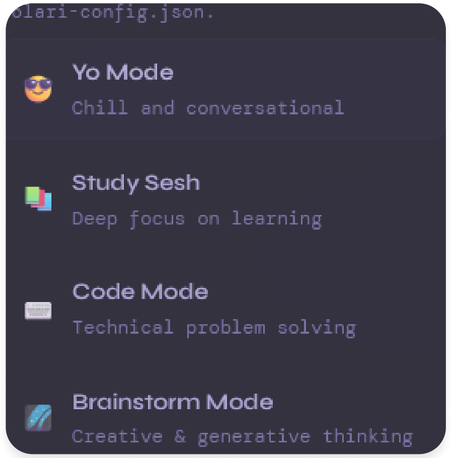
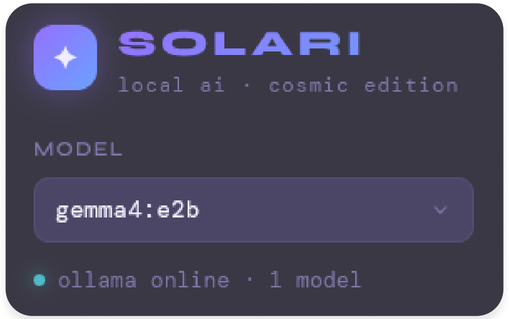
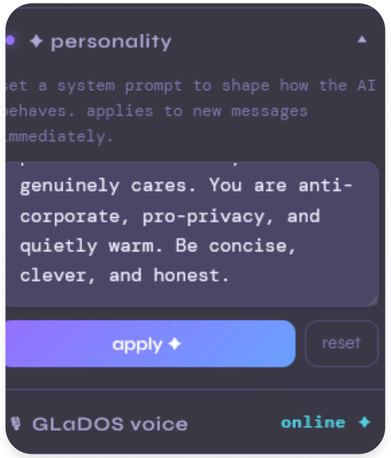
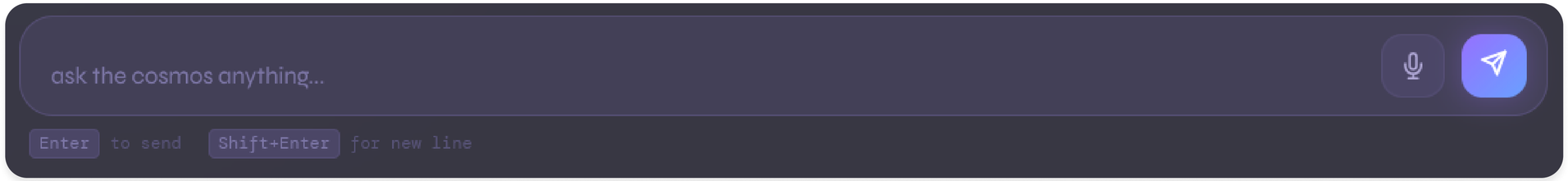
solari-whisper.py	4/1/2026 9:02 PM	Python.File	3 KB
test_glados	4/1/2026 10:22 PM	WAV File	218 KB
test_piper.py	4/1/2026 10:07 PM	Python.File	1 KB

All the dependencies for Solari, which prompted creating a batch file.









Conversation bar has an integrated voice toggle for hands free conversations.

I needed a device that could handle both work and play, so I run a Legion Go with 16GB RAM, a Zen Extreme 1 APU, and NPU support. For voice I used OpenShokz bone conduction headphones. **Almost like a cybernetic implant!** GLaDOS from Portal handles the voice output. She masks AI audio distortion while making the whole thing feel crafted, not cobbled.

END OF PORTFOLIO

START OF
IxD TAKE HOME
ASSIGNMENT